16 EXAM PRACTICE

1 $Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s)$ $\quad E^\ominus = -2.37$ V

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^\ominus = +0.34$ V

What is the standard cell potential, $E^\ominus_{cell}$, of the following cell?

$Mg(s) \mid Mg^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$

A 2.71 V **B** 2.37 V **C** 2.03 V **D** 0.34 V [1]

(Total for Question 1 = 1 mark)

2 $I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq)$ $\quad E^\ominus = +0.54$ V

$2H^+(aq) + O_2(g) + 2e^- \rightleftharpoons H^2O^2(aq)$ $\quad E^\ominus = +0.68$ V

$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightleftharpoons 2H_2O(l)$ $\quad E^\ominus = +1.77$ V

What is observed when a few drops of acidified hydrogen peroxide solution are added to an excess of aqueous potassium iodide?

A The solution turns brown and effervescence occurs.

B The solution turns brown with no effervescence.

C The solution does not change colour and effervescence occurs.

D The solution does not change colour and no effervescence occurs. [1]

(Total for Question 2 = 1 mark)

3 From the following two standard electrode (redox) potentials:

$MnO_2(s) + 4H^+(aq) + 2e^- \rightleftharpoons Mn^{2+}(aq) + 2H_2O(l)$ $\quad E^\ominus = +1.23$ V

$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq)$ $\quad E^\ominus = +1.36$ V

it can be predicted that chloride ions will **not** be oxidised by solid manganese(IV) oxide.

In fact, chlorine can be prepared by heating manganese(IV) oxide with concentrated hydrochloric acid.

Why does this prediction fail?

A The reaction does not produce manganese(II) ions.

B The standard electrode (redox) potentials apply only if one mole of manganese(IV) oxide is used.

C The manganese(IV) oxide needs to be in alkaline solution.

D Standard electrode (redox) potentials apply only under specified conditions of concentration and temperature. [1]

(Total for Question 3 = 1 mark)

4 The standard electrode (redox) potential of four half-cells are listed below:

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^\ominus = +0.34$ V

$AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-(aq)$ $\quad E^\ominus = +0.22$ V

$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$ $\quad E^\ominus = +0.00$ V

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ $\quad E^\ominus = -0.76$ V

Which standard cell potential, $E^\ominus_{cell}$, could be obtained by combining two of these standard half-cells?

A 0.42 V **B** 0.54 V **C** 0.56 V **D** 0.98 V [1]

(Total for Question 4 = 1 mark)

5 The standard electrode potentials of $Ag^+(aq) \mid Ag(s)$ and $Zn^{2+}(aq) \mid Zn(s)$ are +0.80 V and −0.76 V, respectively.

Which conclusion can be made from these data?

A Silver displaces zinc from a solution containing zinc ions.

B Silver is an oxidising agent.

C Zinc has a greater tendency than silver to form positive ions.

D Zinc ions can act as a reducing agent. [1]

(Total for Question 5 = 1 mark)

6 The table shows the formulae of some vanadium ions that can exist in aqueous solution, together with the colour of each ion.

	$VO_3^-(aq)$	$VO^{2+}(aq)$	$V^{3+}(aq)$	$V^{2+}(aq)$
oxidation number of vanadium				
colour of ion	yellow	blue	green	violet

(a) Complete the table by inserting the oxidation numbers. [1]

(b) Iron powder is added to an acidified solution of ammonium vanadate(V), NH_4VO_3, and the mixture is shaken until no further change takes place.

Use the following data to explain what you would expect to observe in the sequence of reactions that take place. Your answer should consider **all** of the electrode reactions listed. [5]

	Electrode reaction	$E^\ominus$/V
1	$V^{2+}(aq) + 2e^- \rightleftharpoons V(s)$	−1.18
2	$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	−0.26
3	$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq)$	+0.34
4	$VO_3^-(aq) + 4H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + 2H_2O(l)$	+1.00
5	$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.44

(Total for Question 6 = 6 marks)

7 (a) Draw a suitably labelled diagram to show how you would measure the standard electrode potential of the $Cu^{2+}(aq) \mid Cu(s)$ half-cell. [6]

(b) The table contains some standard electrode (redox) potentials involving copper and its ions.

	Electrode reaction	$E^{\ominus}$/V
1	$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq)$	+0.15
2	$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34
3	$Cu^+(aq) + e^- \rightleftharpoons Cu(s)$	+0.52

(i) Use the data in the table to explain why the following reaction is likely to occur. [2]

$2Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$

(ii) Use oxidation numbers to explain why this reaction is classified as a disproportionation reaction. [3]

(Total for Question 7 = 11 marks)

8 Chlorine gas can be prepared in the fume cupboard of a laboratory by reacting hydrochloric acid with potassium manganate(VII). The two standard electrode (redox) potentials that relate to this reaction are:

$\frac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq)$ $E^{\ominus} = +1.36\,V$

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$ $E^{\ominus} = +1.52\,V$

(a) State what is meant by the term **standard electrode potential**. [2]

(b) Calculate the standard emf, $E^{\ominus}_{cell}$, for an electrochemical cell constructed from these two redox systems. [1]

(c) (i) Construct an ionic equation for the reaction that takes place between hydrochloric acid and potassium manganate(VII). Include state symbols. [2]

(ii) Use oxidation numbers to explain which species is acting as the reducing agent in this reaction. [2]

(d) When very dilute solutions of potassium manganate(VII) and hydrochloric acid are added together, there is no visible change. Explain, using the changes in electrode potentials that occur, why no reaction takes place when the solutions are diluted. [3]

(Total for Question 8 = 10 marks)

9 Vanadium(V), in the form of vanadate(V) ions, can be reduced in acidified solution by adding sulfur dioxide.

A student carries out an experiment to determine the new oxidation state of the vanadium after the reaction has taken place. He uses this method:

- He dissolves 2.24 g of ammonium vanadate(V), NH_4VO_3, (molar mass = 116.9 g mol^{-1}) in water and makes up the solution to 250 cm^3.
- In a fume cupboard, he bubbles sulfur dioxide through the solution until there is no further colour change.
- He then boils the solution for five minutes and allows the solution to cool.
- He labels this solution '**Solution Y**'.
- He then titrates 25.0 cm^3 of solution Y against 0.0200 mol dm^{-3} $KMnO_4$ solution. He finds that 38.40 cm^3 of $KMnO_4$ are required to oxidise the vanadium.

In this titration:

- The manganate(VII) ions are reduced from the +7 to the +2 oxidation state.
- The vanadium in solution Y is oxidised to the +5 oxidation state.

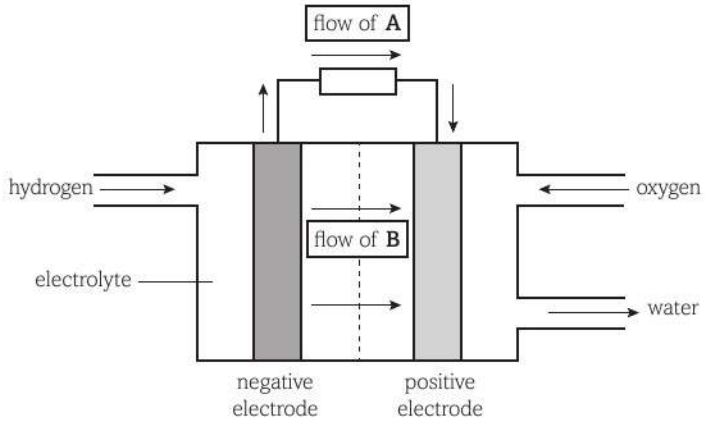
(a) Give a reason why the student carried out the experiment in a fume cupboard. [1]

(b) Use the data to show that the oxidation state of the vanadium in solution Y is +3. [5]

(c) The student repeats the experiment with the same mass of ammonium vanadate(V), but this time he boils the solution for only one minute. He finds that the volume of potassium manganate(VII) solution required in the titration has increased. Explain this observation. [2]

(Total for Question 9 = 8 marks)

10 The diagram shows a hydrogen–oxygen fuel cell.



(a) Identify the particles represented by **A** and **B**. [2]

(b) The two electrodes are coated with a metal catalyst. Name a suitable metal to use as the catalyst. [1]

(c) State two advantages of fuel cells over the use of fossil fuels in cars. [2]

(d) Apart from cost, state two disadvantages of using fuel cells rather than fossil fuels in cars. [2]

(e) The zinc–silver oxide cell is used in button cells for watch batteries. It is based on the following half-cell reactions:

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ $E^{\ominus} = -0.76V$

$Ag_2O(s) + H_2O(l) + 2e^- \rightleftharpoons 2Ag(s) + 2OH^-(aq)$ $E^{\ominus} = +0.34V$

(i) Write the cell diagram for this cell. [2]

(ii) Write the equation for the overall reaction taking place in the cell when it is in use. Include state symbols. [2]

(iii) Calculate the standard emf, $E^{\ominus}_{cell}$, of the cell. [1]

(iv) Use your answer to part (iii) to calculate the equilibrium constant, K, for the reaction at 298 K.

[$R = 8.31\,J\,mol^{-1}\,K^{-1}$] [2]

(Total for Question 10 = 14 marks)